



REVA Academy for Corporate Excellence

Program: MTech/ MSc in Artificial Intelligence	Course: Statistical Foundations
Batch: AI09	Exam: Final
Date: 15 th March 2025	Time: 10.30 am to 1.30 pm
Duration: 3 Hrs	Marks: 100

Instructions

1. This is an open-book exam, but all answers must be paraphrased and customized. Copying or plagiarism will result in suspension and a zero score for identical answers.
2. All questions are mandatory to attend.
3. Answers should be brief and to the point. You may submit a zipped folder with,
 - Word document
 - Well commented scripts, and,
 - PPT/Excel.

Section A: Attempt any **10 questions** from the below.

Instructions	1. Gen AI Tools allowed interpretation and insights from each question is more important than just code. 2. Do not just focus on coding, focus on outcome and practical applications of the outcomes. 3. Attempt any 10 questions, each question carries 10 marks. 4. Find the dataset on your own from the below source. Dataset Source: https://archive.ics.uci.edu/datasets	Marks
Q1.	Auto MPG Dataset (Fuel Efficiency Prediction) Description: This dataset contains information on different car models, including engine displacement, horsepower, weight, acceleration, and miles per gallon (MPG). Linear Regression Questions: a. Develop a linear regression model to predict MPG based on horsepower. Evaluate its accuracy using RMSE and Adjusted R-squared. b. Check whether adding interaction terms (e.g., horsepower $\times$ weight) improves model accuracy.	10



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	c. Conduct a hypothesis test to determine if weight has a significant impact on fuel efficiency. Polynomial Regression Questions: d. Fit a quadratic and cubic regression model to predict MPG based on horsepower. Compare model performance using cross-validation. e. Visualize the polynomial regression curves and discuss whether higher-degree models lead to overfitting	
Q2.	Concrete Strength Dataset Description: This dataset contains information on the compressive strength of concrete based on different material compositions such as cement, water, and fine aggregate. Linear Regression Questions: a. Build a multiple linear regression model to predict concrete strength. Identify which material has the most impact on the strength. b. Analyze the residuals to check for normality and homoscedasticity. c. Use a log-transformed regression model if the relationship between predictors and response variable is non-linear. Polynomial Regression Questions: d. Fit a polynomial regression model to predict strength using cement and water. Evaluate the polynomial degree that minimizes RMSE. e. Compare the AIC/BIC values of linear and polynomial regression models to determine the better-fitting model.	10
Q3.	Bike Sharing Dataset Description: This dataset contains information on the number of bike rentals based on weather, temperature, humidity, and time-related variables. Linear Regression Questions: a. Construct a time-series linear regression model to predict bike rentals based on temperature and humidity.	10



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	b. Include categorical variables such as "weekday" and "holiday" using one-hot encoding in a multiple regression model. Polynomial Regression Questions: c. Fit a polynomial regression model to understand the relationship between bike rentals and temperature. Determine whether a quadratic or cubic model provides a better fit. d. Plot learning curves to examine whether the polynomial regression model is overfitting or underfitting the data.	
Q4.	Energy Efficiency Dataset (Building Energy Consumption Prediction) Description: This dataset contains information on building characteristics, such as wall area, roof area, and glazing area, to predict energy efficiency. Linear Regression Questions: a. Develop a multiple linear regression model to predict heating load based on building characteristics. b. Analyze feature importance using standardized regression coefficients. c. Detect outliers using the Cook's distance method and analyze their impact on the model. Polynomial Regression Questions: d. Fit a polynomial regression model to capture nonlinear relationships between heating load and glazing area. e. Compare linear vs. polynomial models using cross-validation and interpret the results.	10
Q5.	Air Quality Dataset (Pollutant Level Prediction) Description: This dataset contains air pollution data, including CO, NO2, O3, temperature, and humidity. Linear Regression Questions: a. Construct a linear regression model to predict CO concentration based on temperature and humidity.	10



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	b. Perform feature scaling (normalization vs. standardization) and evaluate its impact on model performance. c. Analyze the autocorrelation of residuals to check for time-series effects. Polynomial Regression Questions: d. Fit polynomial regression models of varying degrees to predict pollutant levels. Use AIC/BIC for model selection. e. Examine overfitting by plotting the polynomial regression curve and validation error.	
Q6.	Parkinson's Disease Dataset (Speech Signal Analysis for Diagnosis) Description: This dataset includes voice measurements of individuals, which can be used to predict Parkinson's disease. Linear Regression Questions: a. Develop a linear regression model to predict a health index score based on vocal attributes. b. Identify the most significant vocal features affecting the prediction using p-values and confidence intervals. c. Check for heteroscedasticity using Breusch-Pagan test and propose a remedy if heteroscedasticity is found. Polynomial Regression Questions: d. Fit a polynomial regression model to explore the nonlinear relationship between voice jitter and health index. e. Compare models using RMSE, Adjusted R², and visualize the polynomial curve.	10
Q7.	Power Plant Output Dataset (Energy Generation Prediction) Description: This dataset contains data on temperature, ambient pressure, humidity, and energy output from a power plant. Linear Regression Questions: a. Build a multiple linear regression model to predict power output using environmental factors.	10



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	b. Analyze the correlation matrix to identify multicollinearity issues. c. Perform stepwise regression to select the best predictor subset. Polynomial Regression Questions: d. Fit polynomial regression models to explore nonlinear effects of temperature on power output. e. Determine the optimal degree of polynomial using cross-validation.	
Q8.	Electrical Grid Stability Dataset (Predicting Grid Stability) Description: This dataset contains voltage, current, and power factor values to assess electrical grid stability. Linear Regression Questions: a. Develop a multiple linear regression model to predict grid stability based on voltage fluctuations. b. Perform a residual analysis to check whether the errors are normally distributed. c. Use regularization techniques (Ridge/Lasso) and compare performance with standard linear regression. Polynomial Regression Questions: d. Fit a quadratic regression model to model nonlinear relationships between voltage and stability index. e. Compare the performance of linear vs. polynomial regression using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).	10
Q9.	Student Alcohol Consumption Dataset (Predicting Alcohol Use in High School Students) Description: This dataset contains demographic and lifestyle information about students, along with their alcohol consumption levels. Linear Regression Questions: a. Create a multiple linear regression model to predict weekly alcohol consumption based on age, study time, and family factors.	10



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	b. Assess the impact of parental education on student drinking habits using statistical tests. c. Evaluate the effect of interaction terms (e.g., age $\times$ peer influence) on alcohol consumption. Polynomial Regression Questions: d. Fit a polynomial regression model to examine the non-linear relationship between alcohol consumption and study time. e. Compare R^2 values for different degrees of polynomial regression and interpret the results.	
Q10.	Insurance Charges Dataset (Predicting Medical Insurance Costs) Description: This dataset contains information about insurance policyholders, such as age, BMI, and smoking habits, along with insurance charges. Linear Regression Questions: a. Develop a linear regression model to predict insurance charges based on patient characteristics. b. Identify the most important predictor of insurance cost using feature importance ranking. c. Perform an F-test to check whether at least one predictor variable significantly impacts insurance charges. Polynomial Regression Questions: d. Fit a polynomial regression model to analyze the relationship between BMI and insurance cost. e. Use grid search to optimize the degree of the polynomial regression model.	10
Q11.	Airfoil Self-Noise Dataset (Predicting Noise Levels of Aircraft Wings) Description: This dataset contains different aerodynamic parameters that affect aircraft noise levels. Linear Regression Questions:	10



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	a. Train a multiple linear regression model to predict self-noise levels based on frequency and air velocity. b. Investigate the heteroscedasticity in model residuals using the Breusch-Pagan test. c. Apply log-transformation on the target variable if necessary and analyze its effect on model performance. Polynomial Regression Questions: a. Fit a polynomial regression model to capture the non-linear relationship between frequency and noise level. b. Compare different polynomial models using learning curves to assess underfitting vs. overfitting.	
Q12.	Fish Market Dataset (Predicting Fish Weight) Description: This dataset contains fish species, length, width, and weight measurements. Linear Regression Questions: a. Build a simple linear regression model to predict fish weight using length. b. Extend the model to multiple linear regression by adding species and width as predictors. c. Evaluate model assumptions using QQ-plots and normality tests for residuals. Polynomial Regression Questions: a. Fit a polynomial regression model to study the non-linear relationship between fish weight and length. b. Use L1 and L2 regularization to compare model stability against overfitting.	10

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